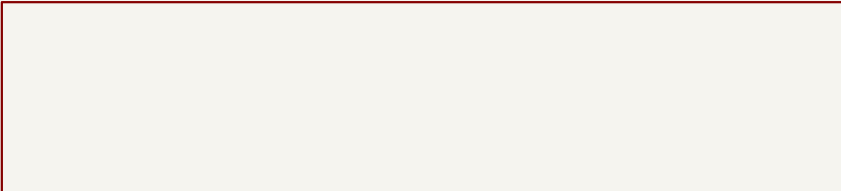
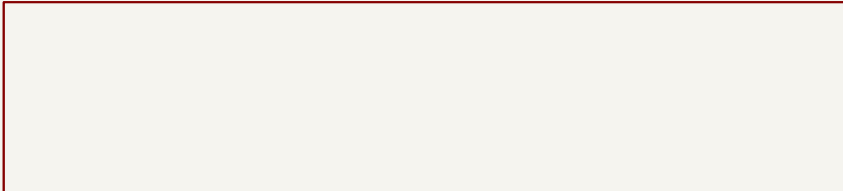
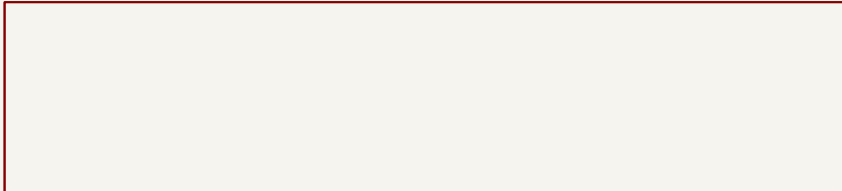
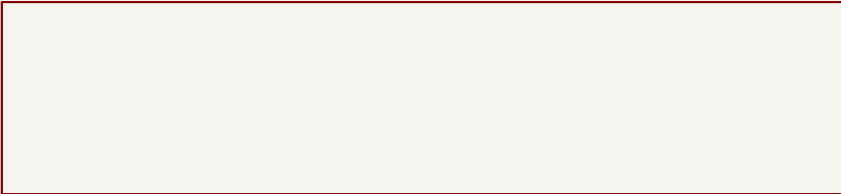
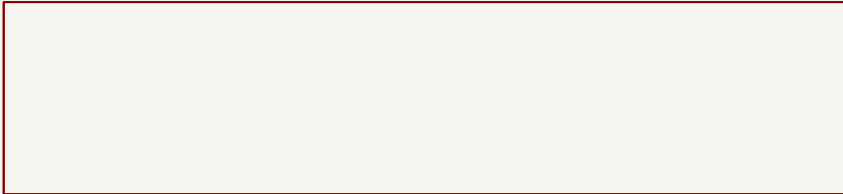
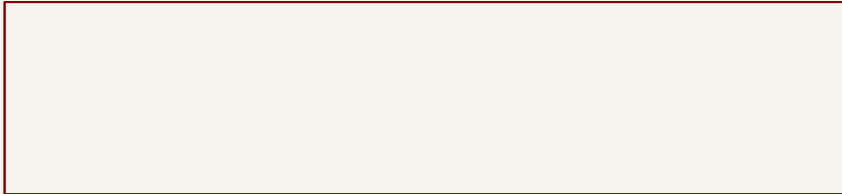
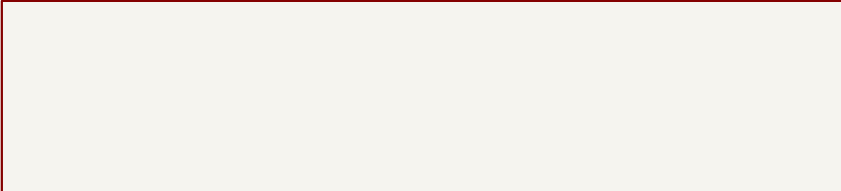
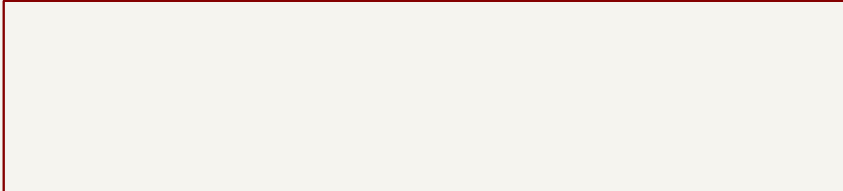
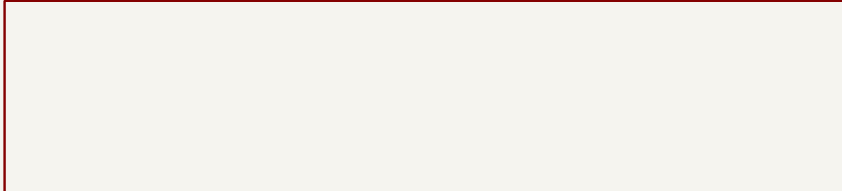
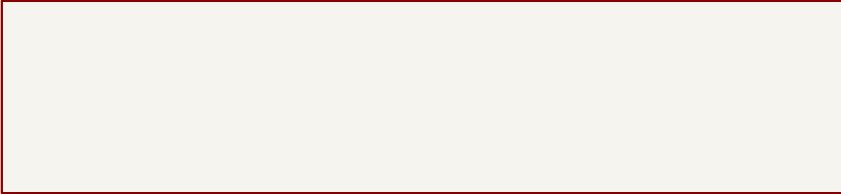
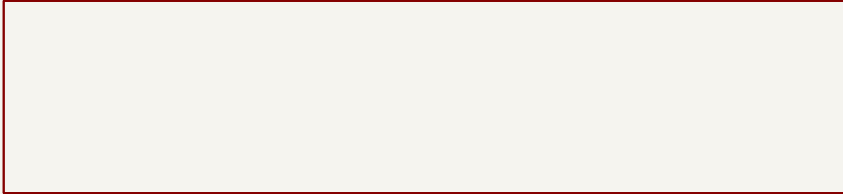
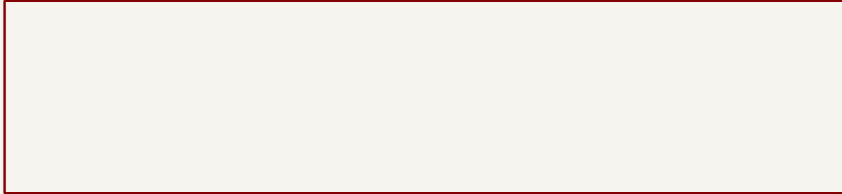
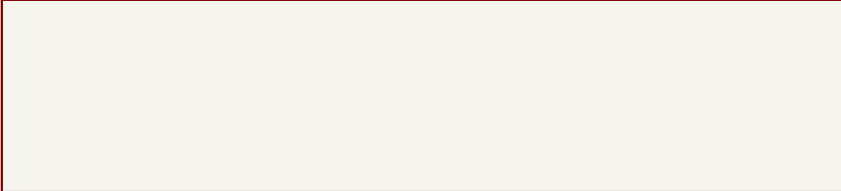


	1	2	3	4	5	6	7	8			
A	AI-GENERATED ENGINEERING PROTOTYPE – V1 – HUMAN REVIEW REQUIRED										
	Design basis: 36–54 V input / 24 V 41.67 A / 100 kHz / 150 MHz F28335 Hardware review prototype. Firmware control loop and thermal performance are unvalidated. All inter-page connections use explicit global net labels. GND is one continuous electrical net.										
B	Power path / half bridge / Kelvin shunt		Input and output bulk capacitor banks		12 V half-bridge gate driver and gate networks						
											
	文件: 01_power.kicad_sch		文件: 02_capacitors.kicad_sch		文件: 03_gate.kicad_sch						
C	48 V to 12 V auxiliary supply		12 V to 5 V auxiliary supply		1.9 V / 3.3 V sequencing and reset						
											
	文件: 04_aux12.kicad_sch		文件: 05_aux5.kicad_sch		文件: 06_dsp_supplies.kicad_sch						
	Current / voltage / temperature measurement		Hardware comparators and fault collection		Latched enable and PWM inhibition						
											
	文件: 07_sensing.kicad_sch		文件: 08_protection.kicad_sch		文件: 09_pwm_logic.kicad_sch						
D	F28335 pin assignment – all 176 pins		DSP per-pin bypass capacitors		Clock / ADC reference / boot / JTAG / UART						
											
	文件: 10_dsp.kicad_sch		文件: 11_dsp_decoupling.kicad_sch		文件: 12_clock_debug.kicad_sch						
E	Test points and mechanical supports										
											
	文件: 13_testpoints.kicad_sch										
F	1	2	3	4	5	6	7	8			

AI-GENERATED ENGINEERING PROTOTYPE – V1 – HUMAN REVIEW REQUIRED

Sheet: /
File: AIPE_Buck_1kW.kicad_sch

Title: 48 V to 24 V / 1 kW synchronous buck

Size: A3

Date:

Rev: V1

KiCad E.D.A. 10.0.7

Id: 1/14

AI-GENERATED ENGINEERING PROTOTYPE – V1 – HUMAN REVIEW REQUIRED

Sheet: / File: AIPE_Buck_1kW.kicad_sch		
Title: 48 V to 24 V / 1 kW synchronous buck		
Size: A3	Date:	Rev: V1
KiCad E.D.A. 10.0.7	Id: 1/14	

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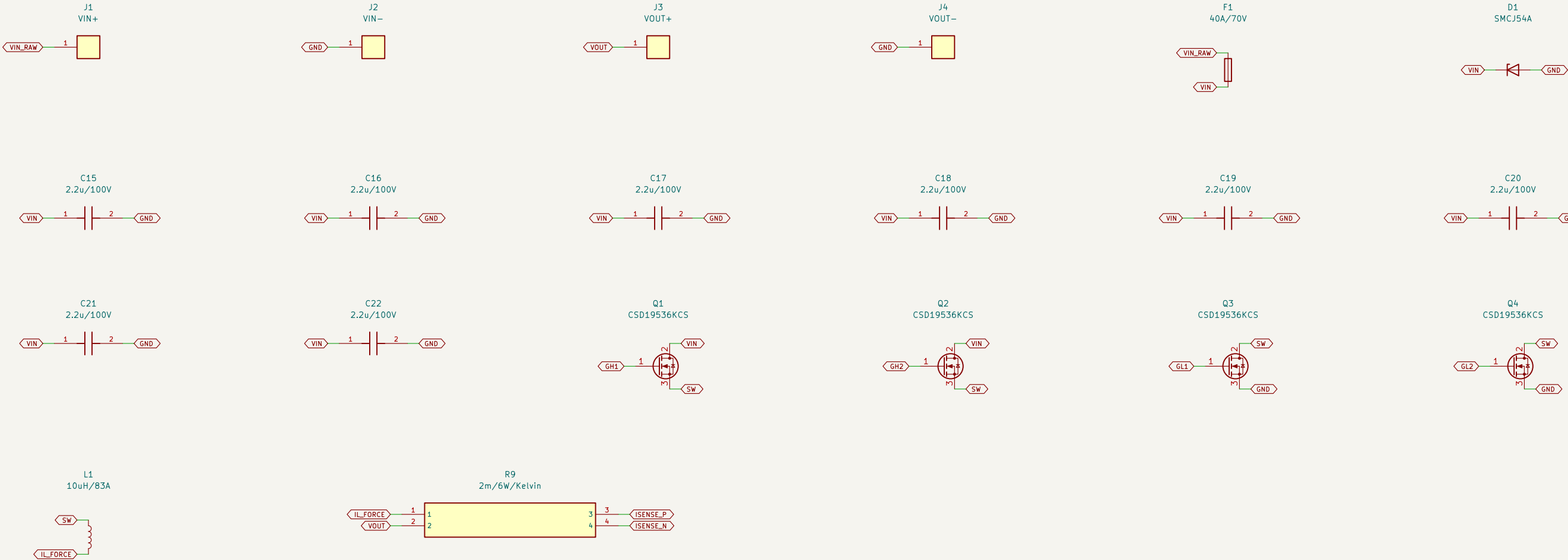
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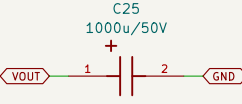
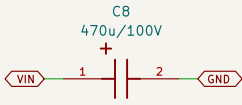
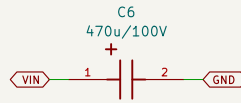
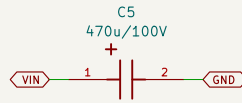
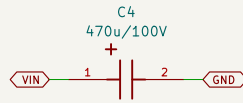
AI-GENERATED ENGINEERING PROTOTYPE – V1 – HUMAN REVIEW REQUIRED

Sheet: /Power path / half bridge / Kelvin shunt/
File: 01_power.kicad_sch

Title: Power path / half bridge / Kelvin shunt

Size: A3	Date:	Rev: V1
KiCad E.D.A. 10.0.7		Id: 2/14

Input and output bulk capacitor banks



AI-GENERATED ENGINEERING PROTOTYPE – V1 – HUMAN REVIEW REQUIRED

Sheet: /Input and output bulk capacitor banks/
File: 02_capacitors.kicad_sch

Title: Input and output bulk capacitor banks

Size: A3	Date:	Rev: V1
KiCad E.D.A. 10.0.7		Id: 3/14

12 V half-bridge gate driver and gate networks

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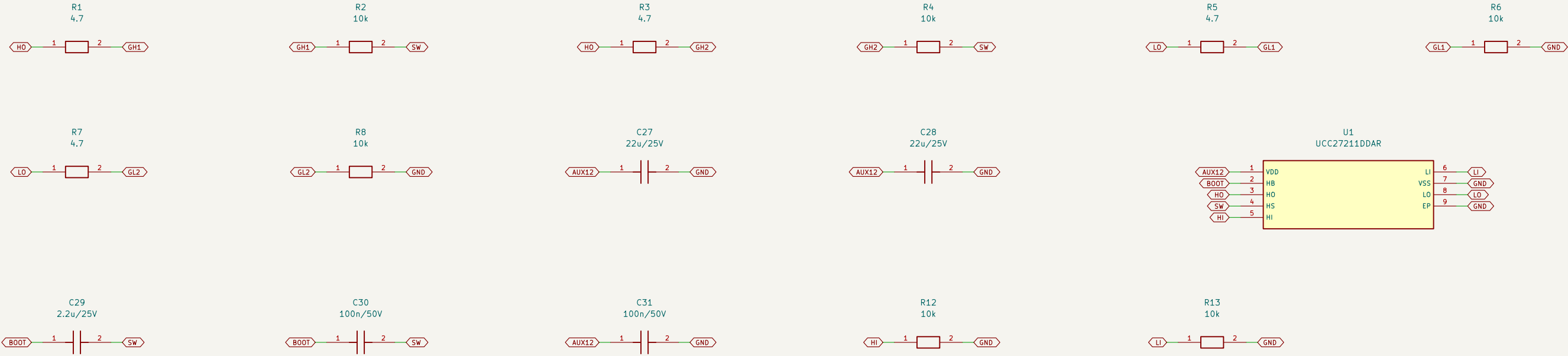
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AI-GENERATED ENGINEERING PROTOTYPE – V1 – HUMAN REVIEW REQUIRED

Sheet: /12 V half-bridge gate driver and gate networks/
File: 03_gate.kicad_sch

Title: 12 V half-bridge gate driver and gate networks

Size: A3	Date:	Rev: V1
KiCad E.D.A. 10.0.7		Id: 4/14

48 V to 12 V auxiliary supply

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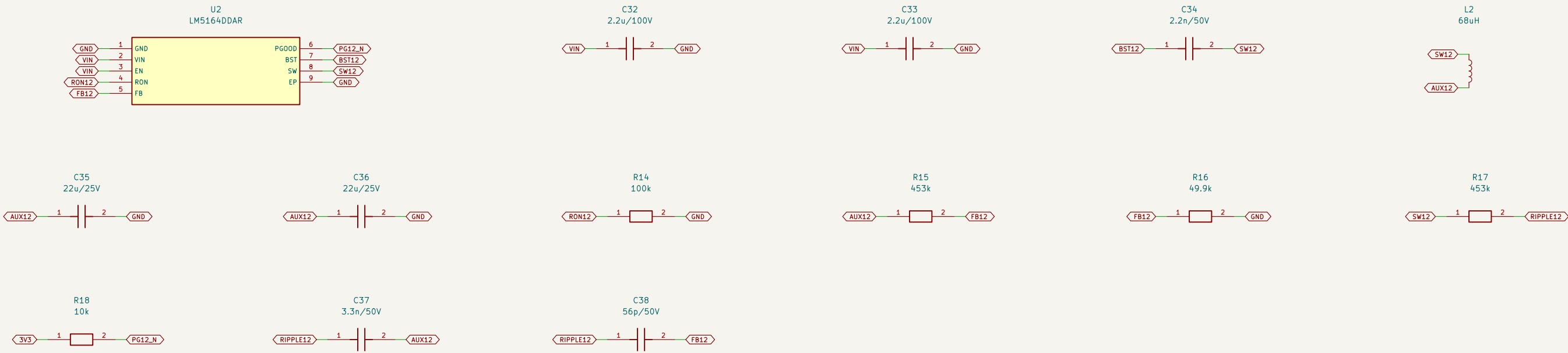
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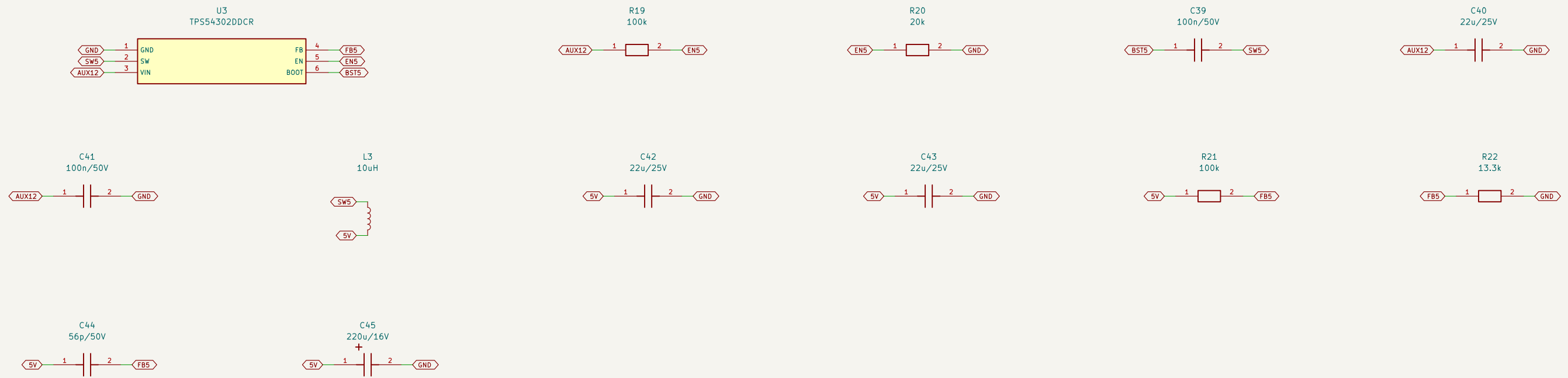
AI-GENERATED ENGINEERING PROTOTYPE – V1 – HUMAN REVIEW REQUIRED

Sheet: /48 V to 12 V auxiliary supply/
File: 04_aux12.kicad_sch

Title: 48 V to 12 V auxiliary supply

Size: A3	Date:	Rev: V1
KiCad E.D.A. 10.0.7		Id: 5/14

12 V to 5 V auxiliary supply



AI-GENERATED ENGINEERING PROTOTYPE – V1 – HUMAN REVIEW REQUIRED

Sheet: /12 V to 5 V auxiliary supply/
File: 05_aux5.kicad_sch

Title: 12 V to 5 V auxiliary supply

Size: A3

Date: _____

Rev: V1

KiCad E.D.A. 10.0.7

Rev. 11
d: 6/14

1.9 V / 3.3 V sequencing and reset

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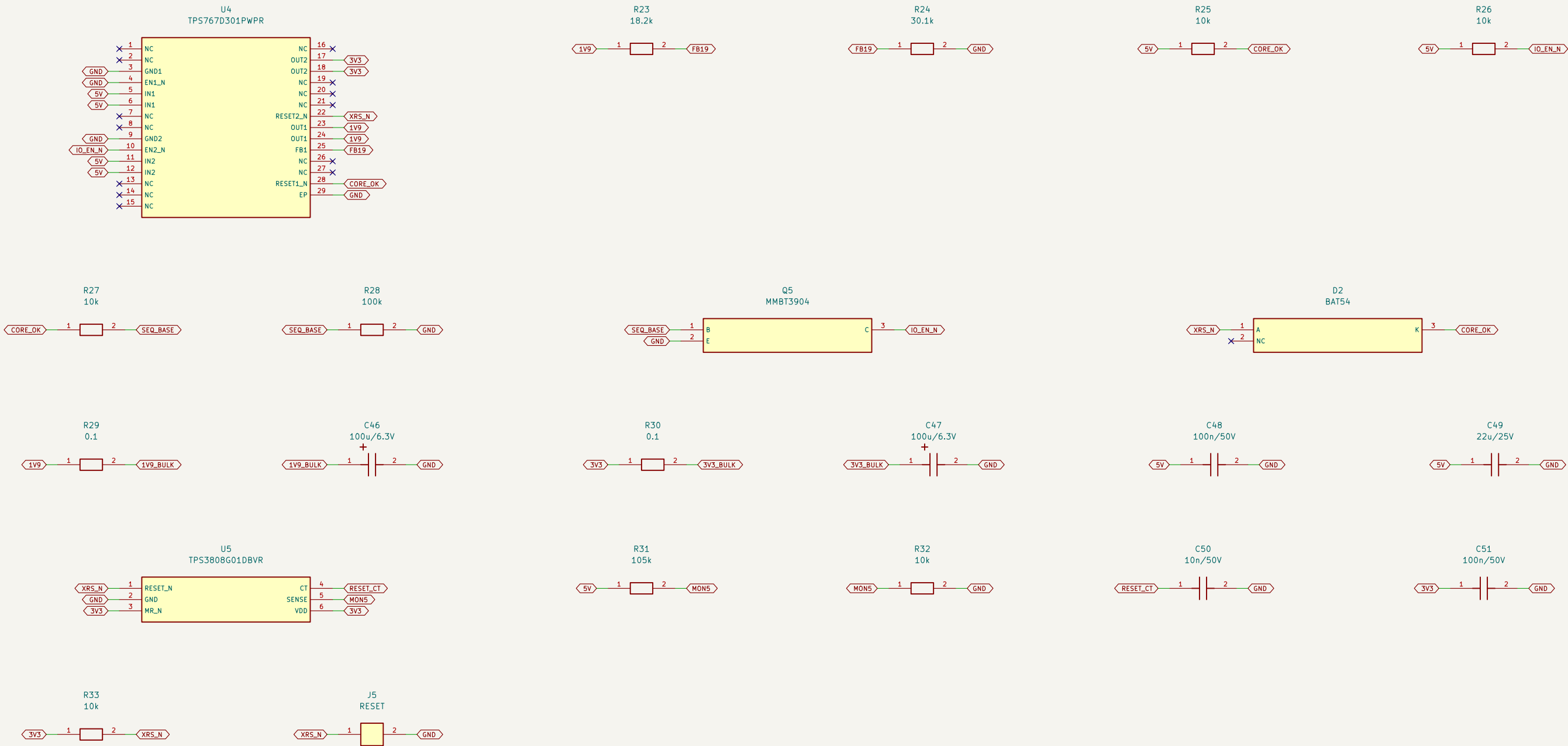
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AI-GENERATED ENGINEERING PROTOTYPE – V1 – HUMAN REVIEW REQUIRED

Sheet: /1.9 V / 3.3 V sequencing and reset/
File: 06_dsp_supplies.kicad_sch

Title: 1.9 V / 3.3 V sequencing and reset

Size: A3

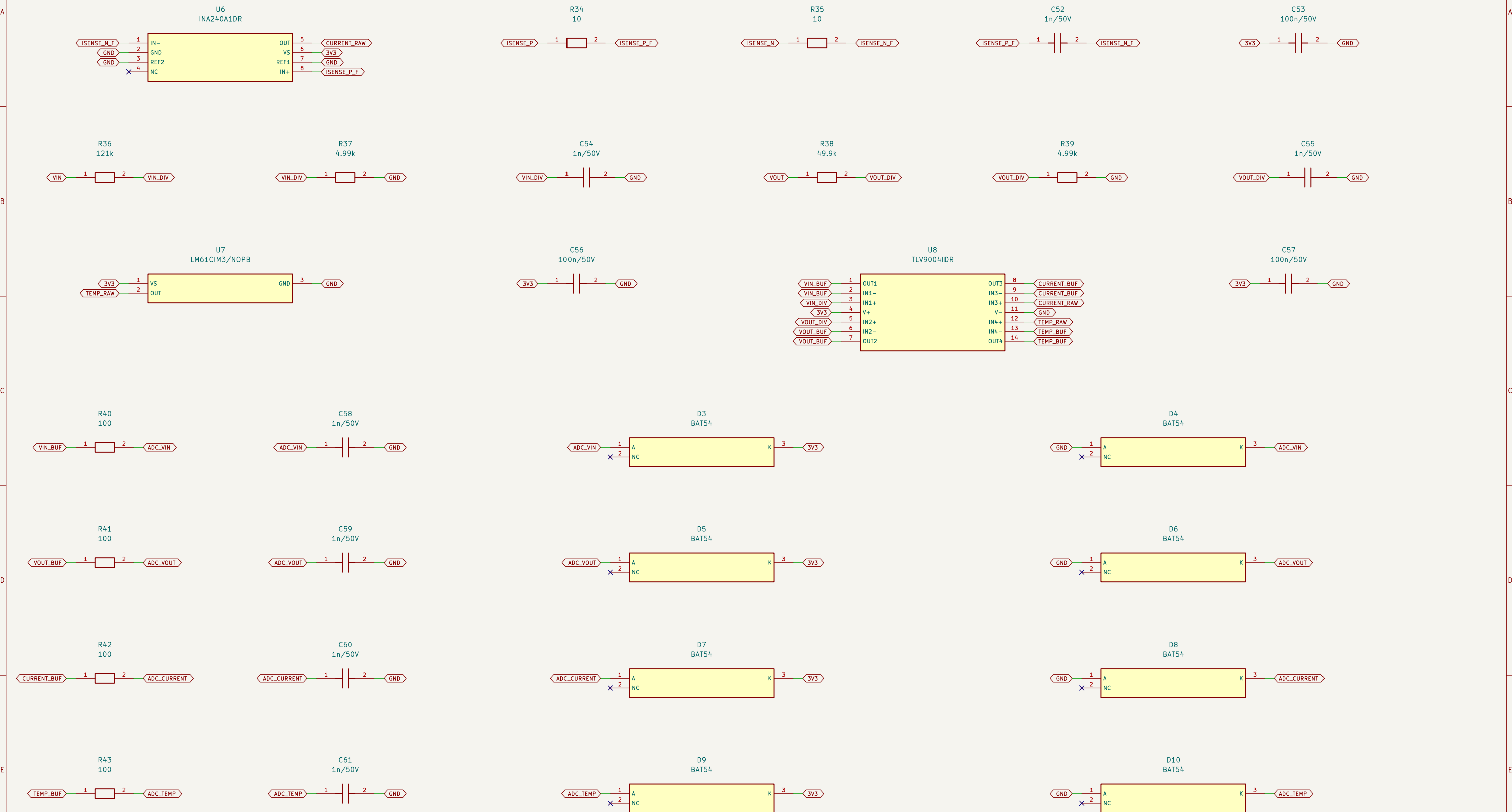
Date:

Rev: V1

KiCad E.D.A. 10.0.7

Id: 7/14

Current / voltage / temperature measurement



AI-GENERATED ENGINEERING PROTOTYPE – V1 – HUMAN REVIEW REQUIRED

Sheet: /Current / voltage / temperature measurement/
File: 07_sensing.kicad_sch

Title: Current / voltage / temperature measurement

Size: A3

Date:

Rev: V1

KiCad E.D.A. 10.0.7

Id: 8/14

Hardware comparators and fault collection

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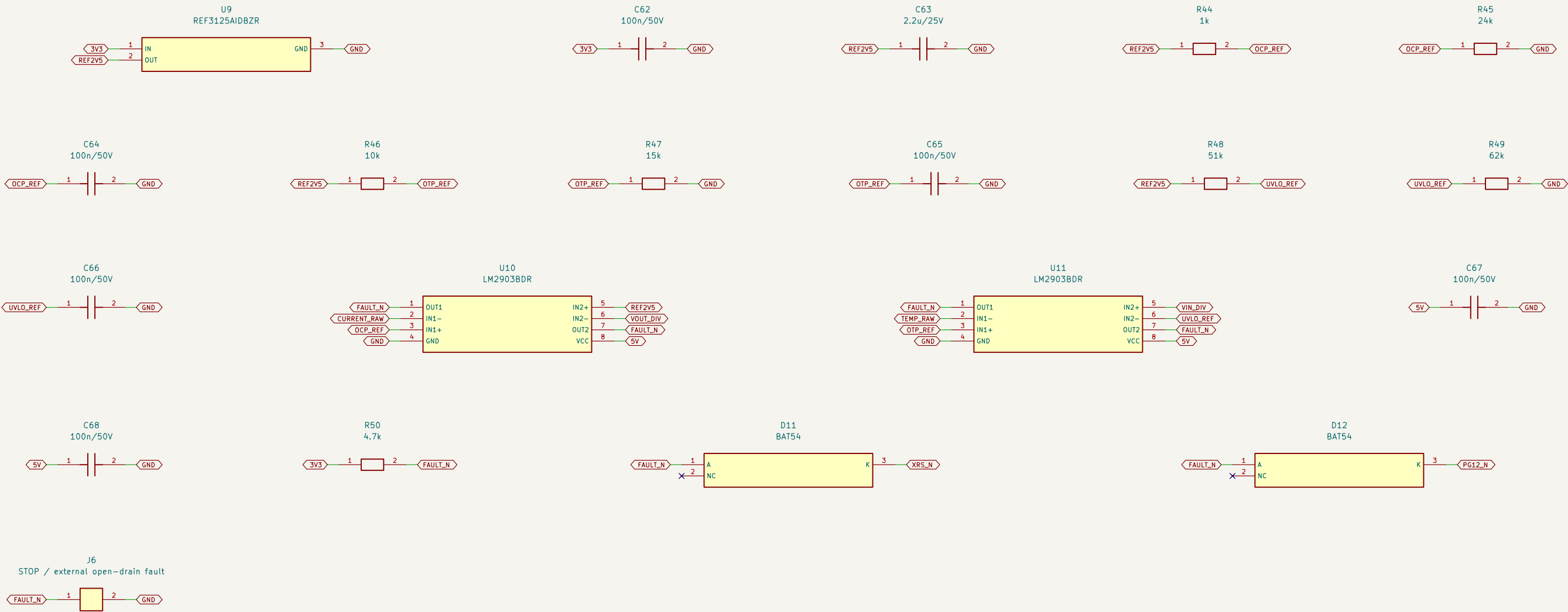
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AI-GENERATED ENGINEERING PROTOTYPE – V1 – HUMAN REVIEW REQUIRED

Sheet: /Hardware comparators and fault collection/
File: 08_protection.kicad_sch

Title: Hardware comparators and fault collection

Size: A3	Date:	Rev: V1
KiCad E.D.A. 10.0.7		Id: 9/14

Latched enable and PWM inhibition

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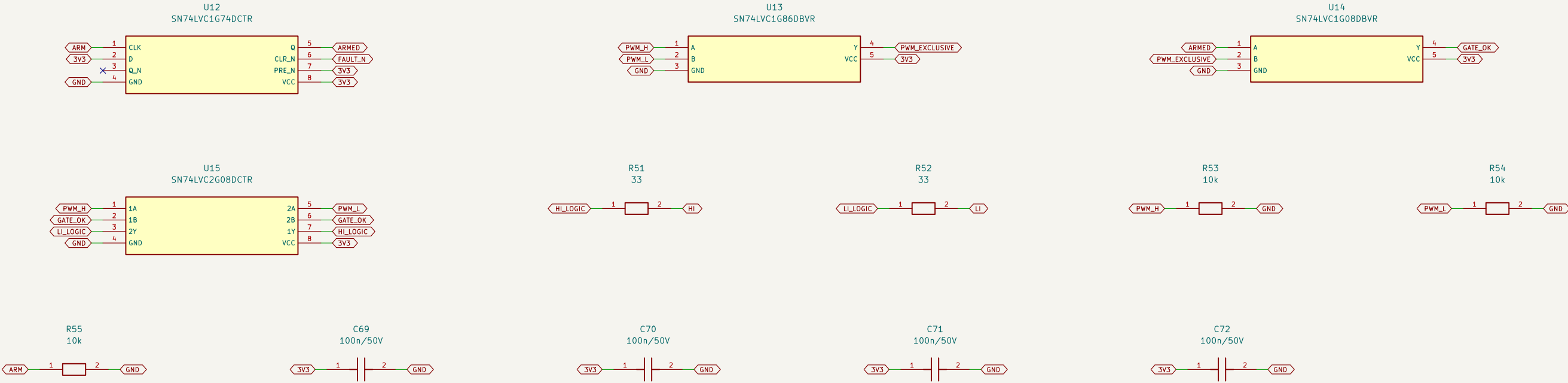
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AI-GENERATED ENGINEERING PROTOTYPE - V1 - HUMAN REVIEW REQUIRED

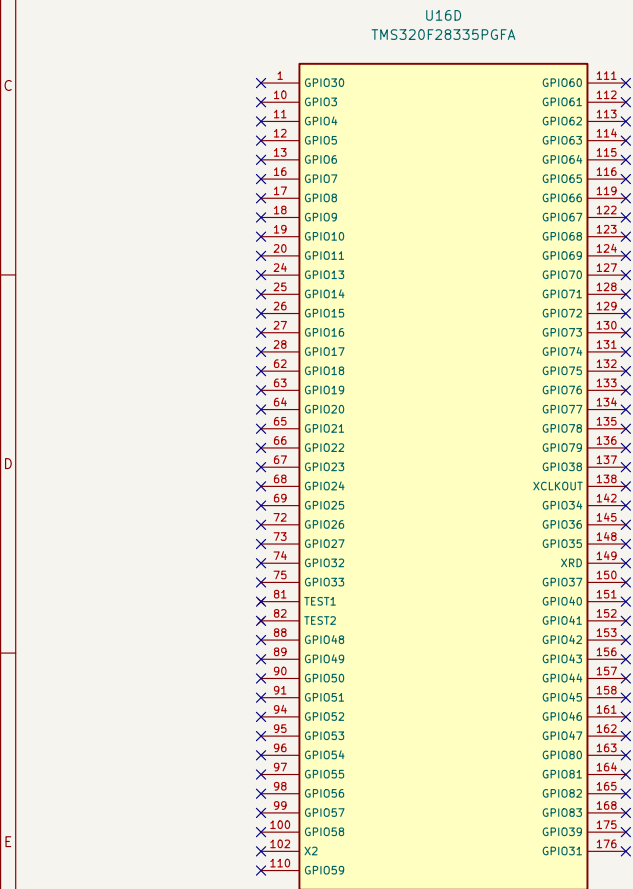
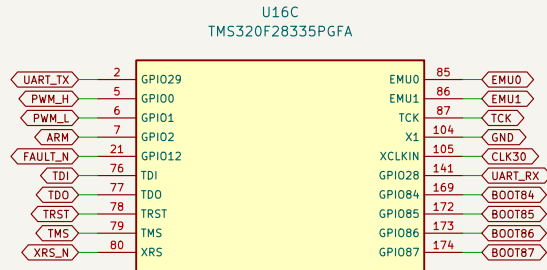
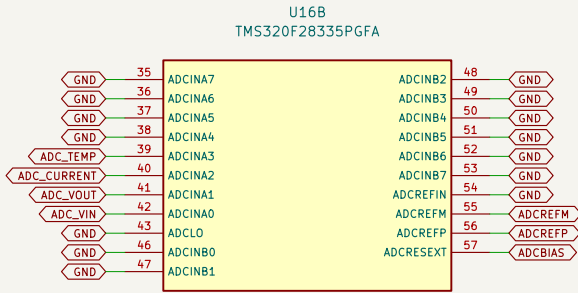
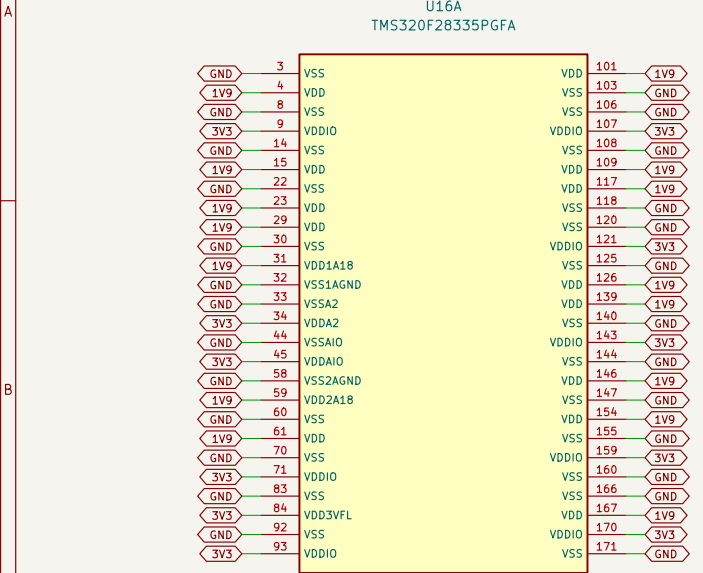
Sheet: /Latched enable and PWM inhibition/
File: 09_pwm_logic.kicad_sch

Title: Latched enable and PWM inhibition

Size: A3	Date:	Rev: V1
KiCad E.D.A. 10.0.7		Id: 10/14

F28335 pin assignment – all 176 pins

Units: supply pins / ADC / connected digital pins / intentionally unused digital pins. TEST1, TEST2 and X2 are NC.



AI-GENERATED ENGINEERING PROTOTYPE – V1 – HUMAN REVIEW REQUIRED

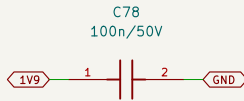
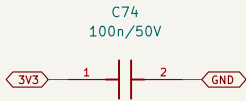
Sheet: /F28335 pin assignment – all 176 pins/
File: 10_dsp.kicad_sch

Title: F28335 pin assignment – all 176 pins

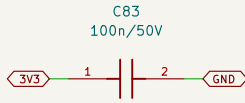
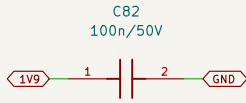
Size: A3	Date:	Rev: V1
KiCad E.D.A. 10.0.7		Id: 11/14

DSP per-pin bypass capacitors

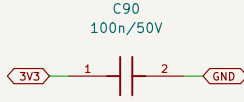
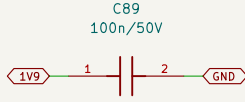
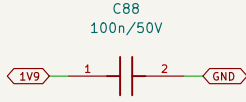
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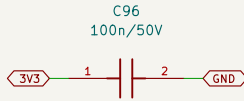
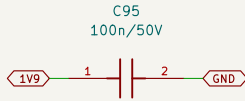
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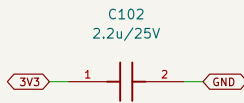
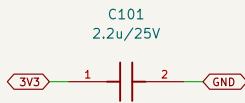
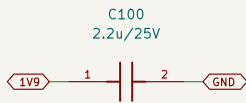
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AI-GENERATED ENGINEERING PROTOTYPE – V1 – HUMAN REVIEW REQUIRED

Sheet: /DSP per-pin bypass capacitors/
File: 11_dsp_decoupling.kicad_sch

Title: DSP per-pin bypass capacitors

Size: A3	Date:	Rev: V1
KiCad E.D.A. 10.0.7		Id: 12/14

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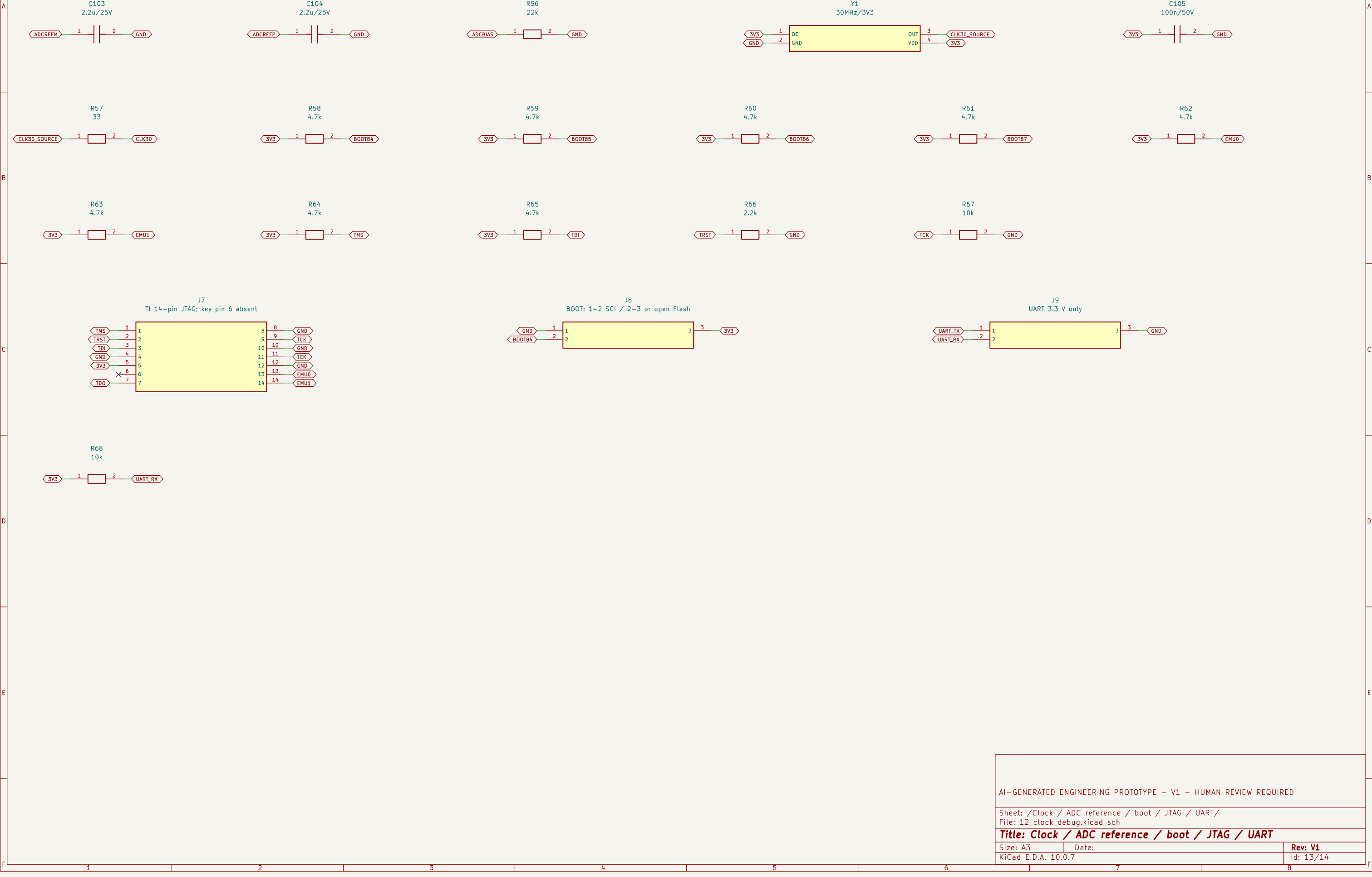
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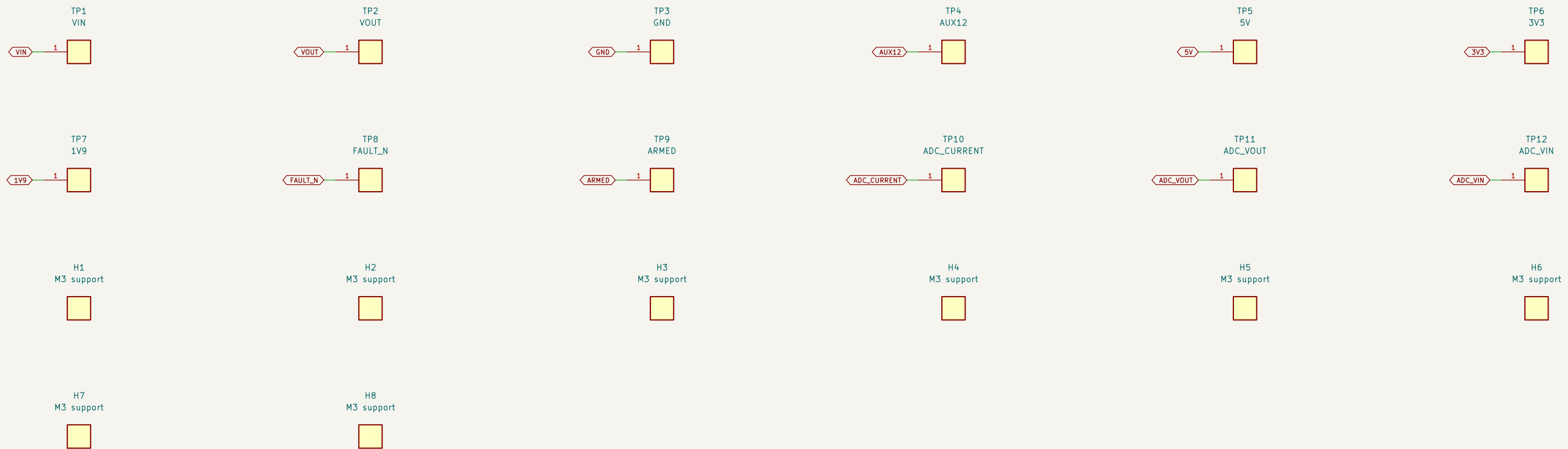
AI-GENERATED ENGINEERING PROTOTYPE - V1 - HUMAN REVIEW REQUIRED

Sheet: /Clock / ADC reference / boot / JTAG / UART/
File: 12_clock_debug.kicad_sch

Title: Clock / ADC reference / boot / JTAG / UART

Size: A3	Date:	Rev: V1
KiCad E.D.A. 10.0.7		Id: 13/14

Test points and mechanical supports



Flags: input source, fuse output, inductor output, auxiliary outputs through inductors, and bootstrap diode supply. They state supply provenance for ERC; they do not prove voltage, current capacity, or protection.

AI-GENERATED ENGINEERING PROTOTYPE – V1 – HUMAN REVIEW REQUIRED

Sheet: /Test points and mechanical supports/
File: 13_testpoints.kicad_sch

Title: Test points and mechanical supports

Size: A3	Date:	Rev: V1
KiCad E.D.A. 10.0.7		Id: 14/14